

The propositions package*

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Abstract

The `propositions` package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like ‘(P)’ or ‘Physicalism’, or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with `amsmath`, `hyperref`, and `cleveref`.

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1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises, . . .) with various kinds of labels. A thesis might be referred to as ‘(P)’ or ‘Physicalism’; the premises of an argument might be numbered as ‘P1’, ‘P2’, ‘P3’, . . . ; or examples might be numbered consecutively over the course of a whole article. Standard $\text{\LaTeX}$ environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The `propositions` package solves this by attaching formatting information to each label. A short item like `\pitem[P]` is displayed as “(P)” and `\ref` automatically produces “(P)” as well—complete with parentheses, hyperlinks, and `cleveref` support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires `amsmath`) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in $\text{\LaTeX}3$, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[options]{propositions}` (see [section 8](#) below for valid package options).

The `prop` environment generates a list of propositions, each introduced by a `\pitem`. `\pitem` with an optional argument gives a `description`-like label:

<code>\begin{prop}</code>	
<code>\pitem[Physicalism] Everything is</code>	Physicalism Everything is physi-
<code>physical. \label{phys}</code>	cal.
<code>\pitem[Idealism] Everything is</code>	Idealism Everything is mental.
<code>mental. \label{ideal}</code>	
<code>\end{prop}</code>	

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

<code>\ref{phys}</code> is more plausible	Physicalism is more plausible than
than <code>\ref{ideal}</code> .	Idealism .

With no optional argument, `\pitem` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

<pre> \begin{prop} \pitem Every atom is physical. \label{atoms} \end{prop} \ref{phys} follows from the conjunction of \ref{atoms} and \begin{prop} \pitem Everything is an atom. \label{atomism} \end{prop} </pre>	(1) Every atom is physical. Physicalism follows from the conjunction of (1) and (2) Everything is an atom.
--	--

As with `enumerate`, the counter and formatting depend on the nesting level:

<pre> \begin{prop} \pitem \label{dual} \begin{prop} \pitem Some things are physical. \label{some} \pitem Some things are not physical. \label{notall} \end{prop} \end{prop} Without \ref{some}, \ref{dual} would be consistent with \ref{ideal}. </pre>	(3) a. Some things are physical. b. Some things are not physical. Without (3a), (3) would be consistent with Idealism.
---	---

4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see [section 5](#) for the possible keys):

<pre> \begin{prop} \pitem[No Overlap, align=flush, display format=\textsc{#1}, ref format=\textit{#1}] Nothing mental is physical. \label{incomp} \end{prop} Is \ref{incomp} consistent with \ref{phys}? </pre>	No OVERLAP Nothing mental is physical. Is <i>No Overlap</i> consistent with Physicalism?
---	---

Preset *item types* can be declared and used instead of configuring the `\pitems` one at a time:

<pre> \begin{prop} \pitem[Nihilism, type=long] There is nothing. \label{nihilism} \end{prop} Does \ref{nihilism} imply \ref{phys}, \ref{dual}, or both? Discuss. </pre>	Nihilism There is nothing. Does Nihilism imply Physicalism, (3), or both? Discuss.
---	---

Shortcut commands—e.g., `\litem[⟨keys⟩]` for `\pitem[type=long, ⟨keys⟩]`—can also be defined. The package loads with a range of predefined types.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\pitems` use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

<code>\begin{equation}</code>	
<code>\exists x (\text{Mental}(x)</code>	
<code>\wedge \text{Physical}(x))</code>	(4) $\exists x(\text{Mental}(x) \wedge \text{Physical}(x))$
<code>\end{equation}</code>	
<code>\begin{prop}</code>	(5) There is overlap between the
<code>\pitem</code>	mental and the physical.
<code>There is overlap between</code>	
<code>the mental and the physical.</code>	
<code>\end{prop}</code>	

The `\ptag` command (requires `amsmath`) is an analogue of `\pitem` that works inside displayed math environments.

<code>\begin{equation}</code>	
<code>\ptag[Monism] \label{mon}</code>	
<code>\exists x \forall y (y = x)</code>	Monism $\exists x \forall y (y = x)$
<code>\end{equation}</code>	
Is <code>\ref{mon}</code> compatible with	Is Monism compatible with (3)?
<code>\ref{dual}</code> ?	

5 The prop environment and \pitem

```
\begin{prop}[\langle keys \rangle]
  \langle environment content \rangle
\end{prop}
```

Creates a displayed list of propositions. It is a standard L^AT_EX list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting package keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper. The optional `\langle keys \rangle` argument accepts the same keys as `\propoptions`^{→ P. 11} (section 8), with effects local to this environment.

```
\begin{inlineprop}[\langle keys \rangle]
  \langle environment content \rangle
\end{inlineprop}
```

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional `\langle keys \rangle` as `prop`.

```
\pitem[\langle keys \rangle]
```

Inside the `prop` and `inlineprop` environments, introduces a labelled proposition. The optional argument is a comma-separated list of `\langle key \rangle = \langle value \rangle` pairs.

When used without an optional argument (or without setting `name`, `counter`, or `type`), the type is determined by the `default nameless type` key (section 8). By default this is `numbered`, which dispatches by nesting depth to `levelone`, `leveltwo`, ... via `\proplevelchoice`.

`\ptag[<keys>]`

(Available only when `amsmath` is loaded.) Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\pitem`^{P.4}, except that `align` has no effect (since positioning is controlled by the tag placement system).

The following keys can be used in the optional argument of `\pitem` or the argument of `\ptag`:

`name=<text>` (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name=<text>`.

`type=<type>` (no default)

An item type, equivalent to a preset collection of keys. Types can be declared using `\SetItemType`^{P.7} or `\DeclareNumberedType`^{P.7}, and several come predefined (section 6).

`counter=<name>` (no default)

Counter to use. The counter is automatically stepped, and the item's `name` is set to `\the<name>` (though this can be overridden by `counter format` or `name`).

`counter format=<template>` (no default)

How to display the counter value. Use `#1` for the counter name, e.g. `counter format=\roman{#1}`. When set, this is used instead of `\the<counter>` for generating the item's name. This allows the same counter to be displayed in different formats by different item types.

`format=<template>` (no default)

Formatting applied to the `name`: use `#1` for the argument, e.g. `format=\textbf{(#1)}`. Shorthand for setting both `display format` and `ref format`.

`display format=<template>` (no default)

Format for displaying the name in the proposition's label. Does not affect cross-references.

`label format=<template>` (no default)

Format applied to the *entire* assembled label, i.e. the name (after `display format`), shorthand, and gloss together. Use `#1` for the whole assembled text. For example, `label format=#1:` appends a colon after the complete label. Does not affect cross-references.

`ref format=<template>` (no default)

Format for subsequent cross-references to this proposition. Does not affect the display.

shorthand= $\langle text \rangle$ (no default)
 An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with **ref format**) rather than the full name.

shorthand format= $\langle template \rangle$ (initially $\sim \text{[#1]}$)
 Format for displaying the shorthand in the label.

gloss= $\langle text \rangle$ (no default)
 A parenthetical gloss displayed after the name. Does not affect cross-references.

gloss format= $\langle template \rangle$ (initially $\sim \text{(#1)}$)
 Format for displaying the gloss in the label.

ref= $\langle text \rangle$ (no default)
 Explicitly set the reference text, overriding what would be derived from **name**, **counter**, or **shorthand**.

label= $\langle label \rangle$ (no default)
 Equivalent to a trailing `\label{\langle label \rangle}`.

crefname= $\langle type \rangle$ (no default)
 When **cleveref** is loaded, assigns an arbitrary reference type to this proposition. For example, **crefname=lemma** causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default **proposition** type. The $\langle type \rangle$ must be known to **cleveref**; new types can be declared with `\crefname`.

align= $\langle type \rangle$ (no default)
 How the label should be positioned. Possible values: **left** (standard left-aligned label, offset controlled by `\labelwidth` and `\labelsep`), **right** (right-aligned, like **enumerate**), **center** (centered within the label box), **flush** (aligned with left margin of item text), **nextline** (label on its own line), and **flush-nextline**. Has no effect inside `inlineprop` or `\ptag`. For left-margin alignment, use **labelindent=0em** instead.

labelindent= $\langle length \rangle$ (no default, per-item)
 Positions the label's left edge at $\langle length \rangle$ from the enclosing left margin. For example, **labelindent=0em** aligns the label with the surrounding text. When the label plus `\labelsep` would overflow into the item text, the first line is pushed right automatically (like the global **labelindent** key). Can be set per-item, per-type, or globally via `\proptoptions`.

labelwidth= $\langle length \rangle$ (no default, per-item)
 Overrides the effective label box width for this item. The label is repositioned as if `\labelwidth` were $\langle length \rangle$, without changing the actual list dimension.

labelsep= $\langle length \rangle$ (no default, per-item)
 Overrides the effective label-to-text gap for this item.

itemindent= $\langle length \rangle$ (no default, per-item)
 Overrides the effective first-line indent for this item. Only increases are fully supported; decreases are clamped to zero.

6 Item types

\SetItemType{ $\langle name \rangle$ }{ $\langle keys \rangle$ }

Defines or modifies an item type for use with the **type** key. All **\pitem**^{P.4} keys are accepted, plus the following:

type= $\langle parent \rangle$ (no default)

Inherit from a parent type. When an item is created, the parent's settings are loaded first (recursively, if the parent itself has a parent), then this type's own keys are applied on top.

The argument may be a macro which is expanded when the item is created: for example, a type with **type**= $\{\langle type1 \rangle\}, \{\langle type2 \rangle\}, \{\langle type3 \rangle\}$ will resolve to $\{\langle type1 \rangle\}, \{\langle type2 \rangle\}, \{\langle type2 \rangle\}$ depending on the nesting depth. The built-in **numbered** and **eqnum** types use this mechanism.

macro= $\langle command \rangle$ (no default)

A new user macro, equivalent to **\pitem**[**type**= $\langle name \rangle$]. Any further keys given to the macro are passed to **\pitem**.

If the type $\langle name \rangle$ already exists, **\SetItemType** modifies or adds keys. For example, **\SetItemType**{**default**}{**align**=**flush**} changes the alignment of the built-in **default** type while preserving its other settings.

<pre> \SetItemType{angle}{ labelindent = 0em, display format = \textbf{\langle angle \rangle}, ref format = \langle angle \rangle, macro = \angitem } \begin{prop} \angitem[Angle thesis] Everything is angular. \end{prop} No further discussion of \lastref{} is needed.</pre>	\langle Angle thesis \rangle Everything is angular. No further discussion of \langle Angle thesis \rangle is needed.
--	--

\DeclareNumberedType{ $\langle name \rangle$ }[$\langle keys \rangle$]

Creates a new L^AT_EX counter named $\langle name \rangle$ and a matching item type with **counter**= $\langle name \rangle$. All **\SetItemType** keys are accepted, plus:

parent= $\langle counter \rangle$ (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as **\numberwithin**). A dedicated **prop** counter (stepped by each **prop** and **inlineprop**) is available for non-persistent numbering.

<code>\DeclareNumberedType{P}</code>	P0 First premise.
<code>\begin{prop}</code>	
<code>\pitem[counter=P] First premise.</code>	P0 Second premise.
<code>\label{p1}</code>	From P0 and P0 ...
<code>\pitem[counter=P] Second premise.</code>	
<code>\label{p2}</code>	
<code>\end{prop}</code>	
From <code>\ref{p1}</code> and <code>\ref{p2}\ldots</code>	

6.1 Built-in types

The following item types are predefined. `\SetItemType`^{P.7} can modify their behaviour.

Type	Shortcut	Counter	Display	Ref	Align
plain	none	none	Name	Name	left
default	none*	none	Name	Name	left
long	<code>\litem</code>	none	Name	Name	flush
bullet	<code>\bitem</code>	none	$\bullet$ / $-$ / $*$ / $\diamond$	(same)	default
roman	<code>\ritem</code>	roman	(i)	(i)	left
alph	<code>\aitem</code>	alph	(a)	(a)	left
levelone	none	numpropi	(1)	(1)	left
leveltwo	none	numpropii	a.	(1a)	left
levelthree	none	numpropiii	(i)	(1a.i)	left
levelfour	none	numpropiv	A.	(1a.i A)	left
levelfive	none	numpropv	(I)	(1a.i A.I)	left
equation	none	equation	(1)	(1)	left
numbered [†]	none	dispatches to levelone...levelfive by depth			
eqnum [†]	none	dispatches to equation, leveltwo...levelfive by depth			

* The default type is auto-selected when `\pitem` or `\ptag` has a **name** or **counter** but no **type** key.

† The **numbered** and **eqnum** types use `type = \proplevelchoice{...}` to dispatch to the appropriate level-specific type at resolution time. **numbered** is the default nameless type; the **equations** package option changes it to **eqnum**.

◇ The **bullet** type uses `\proplevelchoice` to vary the symbol by nesting depth, matching standard `\itemize` symbols.

The **numpropii**–**numpropv** counters reset automatically when a `\pitem` at the next lower level is processed.

`\proplevelchoice{<item1, item2, ...>}`

Picks the entry at the current nesting depth (1-indexed, clamped to the list length). At depth 0 (outside any **prop** environment), returns the first entry. This macro is fully expandable and can be used in the **name**, **format**, and **type** keys of `\SetItemType` to create depth-dependent types.

7 Cross-referencing

Labels placed after `\pitem` items work with the standard `\label/\ref` mechanism. The key difference from ordinary $\text{\LaTeX}$ references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

```
\Ref{\label{}}
\Ref*{\label{}}
```

Titlecase variants of `\ref` and `\ref*`: uppercases the first letter of the formatted output. (For example, if `\ref{thesis}` produces ‘the Identity Theory’, then `\Ref{thesis}` produces ‘The Identity Theory’.) The starred form suppresses the hyperlink. Note: `\Ref` only affects references produced by this package (which are stored via `\propapply`); for other references, it behaves like `\ref`.

```
\nref{\label{}}
\nref*{\label{}}
\Nref{\label{}}
\Nref*{\label{}}
```

“Naked ref.” Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces ‘(P1)’, then `\nref{premise}` produces ‘P1’. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` is often useful in the argument of `\pitem`, when the the name of one proposition should depend on that of another:

<pre>\SetItemType{default}{format=(#1)} \begin{prop} \pitem[Phys] Everything is physical. \label{phys2} \pitem[\nref{phys2}*] Almost everything is physical. \label{newphys2} \pitem[\ref{premise}*] Note that the reference in the name of proposition \ref{newphys2} is made using \ref instead of \nref. \end{prop}</pre>	<pre>(Phys) Everything is physical. (Phys*) Almost everything is physical. (7*) Note that the reference in the name of proposition (Phys*) is made using \nref: using \ref instead would have produced two sets of parentheses, one from the reference and one applied by the display format.</pre>
--	---

Warning: documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple $\text{\LaTeX}$ runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

```
\oref[<prefix>][<suffix>]{<label>}
\oref*[<prefix>][<suffix>]{<label>}
\Oref[<prefix>][<suffix>]{<label>}
\Oref*[<prefix>][<suffix>]{<label>}
```

“Ref with options.” Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, `<suffix>` is appended;

with two, $\langle prefix \rangle$ is prepended and $\langle suffix \rangle$ appended. For instance, if $\backslash\text{ref}\{\text{premise}\}$ produces ‘(P1)’, then $\backslash\text{oref}[*]\{\text{premise}\}$ produces ‘(P1*)’ and $\backslash\text{oref}[\text{old}\sim][*]\{\text{premise}\}$ produces ‘(old~P1*)’.

$\backslash\text{Oref}$ is the titlecase variant; the starred forms suppress the hyperlink.

$\backslash\text{oref}$ can also be useful in the name of $\backslash\text{pitems}$, if one wants the display format for the modified item to depend on that originally used

```
\begin{prop}
  \label{premise}
  \pitem[type=plain,name=\oref[*]{phys2}]
  This will use boldface and
  parentheses because the original
  referenced item did.
\end{prop}
```

(Phys*) This will use boldface and parentheses because the original referenced item did.

Another handy use for $\backslash\text{oref}$ is in combination with $\backslash\text{nref}$ to refer to ranges:

```
The first two numbered examples in
this document were
were \oref[--\nref{atomism}]{atoms}.
```

The first two numbered examples in this document were (1–2).

$\backslash\text{lastref}[\langle prefix \rangle]\{\langle suffix \rangle\}$

$\backslash\text{Lastref}[\langle prefix \rangle]\{\langle suffix \rangle\}$

Formatted reference to the most recently processed $\backslash\text{pitem}$ or $\backslash\text{ptag}$, even without a $\backslash\text{label}$. Useful for back-references in running text. With one argument, $\langle suffix \rangle$ is appended; with two, $\langle prefix \rangle$ is also prepended. Use $\backslash\text{lastref}\{\}$ for a plain reference. $\backslash\text{Lastref}$ is the titlecase variant: it uppercases the first letter.

$\backslash\text{nLastref}$

Like $\backslash\text{lastref}\{\}$ but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

$\backslash\text{parentref}[\langle prefix \rangle]\{\langle suffix \rangle\}$

$\backslash\text{Parentref}[\langle prefix \rangle]\{\langle suffix \rangle\}$

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as $\backslash\text{lastref}$. $\backslash\text{Parentref}$ is the titlecase variant: it uppercases the first letter.

$\backslash\text{nParentref}$

Like $\backslash\text{parentref}\{\}$ but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards.

$\backslash\text{parentref}$ and $\backslash\text{nParentref}$ are useful for making subitems whose names derive from their parent’s:

<pre> \DeclareNumberedType{inner}[counter format=\alph{inner}, display format=#1.] \begin{prop} \pitem[OI] Outer item. \label{outer2} \begin{prop} \pitem[type=inner, format=\parentref{.#1}] \label{dsub1} Ref: \ref{dsub1}, naked: \nref{dsub1}. \pitem[type=inner, display format=#1., ref format=\parentref{.#1}] \label{dsub2} Ref: \ref{dsub2}, naked: \nref{dsub2}. \end{prop} \end{prop} </pre>	(OI) Outer item. (OI.a) Ref: (OI.a), naked: a. b. Ref: (OI.b), naked: b.
--	---

The built-in `leveltwo` and `levelthree` types have `ref format=\parentref{#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as ‘(P1)’, a `leveltwo` sub-item references as ‘(P1a)’ and `\nref` returns just ‘a’.

7.1 How it works: `\propapply`

`\propapply{<template>}{<content>}`

Internally, each reference is stored in the `.aux` file as `\propapply{<template>}{<content>}`.

The `<template>` contains formatting with the placeholder `\propfmtarg` where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to `<content>`. The `\oref`^{P.9} and `\nref`^{P.9} commands work by locally redefining `\propapply`.

In normal use, you need not interact with `\propapply` directly.

`\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`.

8 Package options

`\proppositions{<keys>}`

Sets package-level keys. These can also be set:

- In the optional argument of `prop` and `inlineprop` (local to that environment).
- In the preamble with `\usepackage[<options>]{propositions}` (global).

Exception 1: keys containing # (such as `equation format`) cannot be set in the optional argument of `\usepackage`, due to how L^AT_EX handles # in option values.

Exception 2: `equations` is a global key; it cannot be used in the optional argument of `prop` or `inlineprop`.

8.1 List dimensions

`topsep`= $\langle length / length list \rangle$
`partopsep`= $\langle length / length list \rangle$
`itemsep`= $\langle length / length list \rangle$
`parsep`= $\langle length / length list \rangle$
`leftmargin`= $\langle length / length list \rangle$
`rightmargin`= $\langle length / length list \rangle$
`labelwidth`= $\langle length / length list \rangle$
`labelsep`= $\langle length / length list \rangle$
`itemindent`= $\langle length / length list \rangle$
`listparindent`= $\langle length / length list \rangle$

Override the standard L^AT_EX list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent`= $\langle length / length list \rangle$ (no default)

Positions label left edges at $\langle length \rangle$ from the enclosing margin, adjusting `\labelsep` or `\itemindent` as needed. Positive values move rightward, negative leftward.

`tightspacing` (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (`topsep` and `itemsep` to 2pt with stretch/shrink, `parsep` to 0pt, `partopsep` to 1pt). Individual dimension keys set afterward override.

`nosep` (no value)

Sets `\topsep`, `\itemsep`, and `\parsep` all to zero.

`continue`= $\langle boolean \rangle$ (initially `true`)

When `true` (the default), a `prop` environment that immediately follows another `prop` (with no intervening paragraph text) replaces the normal `\topsep`-based inter-list space with `\itemsep + \parsep`, giving the visual appearance of a single continuous list. This works at any nesting level. Set to `false` to suppress this behaviour for a specific environment or globally via `\propoptions`^{→ P. 11}.

8.2 Default types

When `\pitem` or `\ptag` is used without an explicit `type`, one of the following defaults is selected based on whether a name/counter was given and whether `ptag` mode is active.

`default type`= $\langle type \rangle$ (initially default)

The type used when a `name` or `counter` is given but no explicit `type`.

`default ptag type=<type>` (initially empty)

If set, overrides `default type` for `\ptag` items only.

`default nameless type=<type>` (initially `numbered`)

The type used when `\pitem` or `\ptag` has no optional argument (no name, counter, or type). The built-in `numbered` type dispatches to `levelone...levelfive` by nesting depth via `\proplevelchoice` (see [subsection 6.1](#)).

`default ptag nameless type=<type>` (initially empty)

If set, overrides `default nameless type` for `\ptag` items only.

8.3 Formatting and referencing equation numbers

`equations` (no value, **global only**)

An alias for `default nameless type=eqnum`. The built-in `eqnum` type is like `numbered` but uses the `equation` counter at nesting level 1, sharing counter values with standard displayed equations. Also installs the equation format hooks (see below), so that `\oref` and `\nref` work with equation labels.

`equation format=<template>` (initially `#1`)

Shorthand: sets both `equation display format` and `equation ref format`. Also installs the equation format hooks independently of the `equations` option.

`equation display format=<template>` (initially `#1`)

Controls how equation tags appear in the PDF (via `\tagform@`). Use `#1` for the number. Does not affect `\ref` output. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

`equation ref format=<template>` (initially `#1`)

Controls how `\ref` (and `\oref`, `\nref`) render equation labels. Use `#1` for the number. Does not affect the displayed tag. Also installs the equation format hooks independently of the `equations` option. Locally scoped.

9 Compatibility

The `propositions` package is designed to work with `hyperref`, `cleveref`, and `amsmath`. `amsmath` is required for `\ptag` and the `equations` option. With `cleveref` loaded, all `\pitems` are assigned to a default `proposition` reference type; the `crefname` key can override this.

Recommended load order:

```
\usepackage{hyperref}
\usepackage{amsmath} % if using \ptag
\usepackage{propositions}
\usepackage{cleveref} % if used
```

10 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an `ams-math` equation environment, `hyperref` may emit warnings of the form:

```
pdfTeX warning: destination with the same  
identifier (name{equation.N}) has been already  
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.